



PCI AI

| Project Controls
Institute Global, Inc.

PCI AI · FORMULA SHEET

The Project Management Formula Sheet

Every formula a delivery leader is accountable for
— including the ones only PCI publishes

PERT · CPM · EMV · E[wait] · WSJF · PTA

What this is

Forty-six formulas across project and programme delivery: business case, estimating, critical path, earned value, risk, agile and flow, procurement, quality, stakeholders and portfolio.

Eight of them exist nowhere else. The **governance arithmetic** in section 04 — decision latency, gate net value, committee capacity, baseline drift — is original to the PML-AI Body of Knowledge, because governance is the part of delivery everyone manages by instinct and nobody prices.

Save this one. It is the reference, not a highlight reel.

CONVENTIONS

One rule before the rest

Under hybrid delivery, velocity forecasting and earned value answer the same question through different instruments.

State which one governs the commitment **before** you report either.

And rates and periods must agree. An annual rate against monthly periods is the arithmetic error the verification suite catches most often across the whole programme.

01

Business case and benefits.

Domains 2 and 16. Why the project exists, and whether it delivered.



PCI AI

| Project Controls
Institute Global, Inc.

Appraising the case

EXHIBIT 1

FORMULA	MEANING	DOMAIN
$NPV = \sum CF_t \div (1 + r)^t - I_0$	Net present value	2.3
$IRR: NPV = 0$	Internal rate of return	2.3
$ROI = (Benefit - Cost) \div Cost$	Return on investment	2.3
$BCR = PV(benefits) \div PV(costs)$	Benefit-cost ratio	2.3
Payback	Period to recover the outlay	2.3

Discount both sides or neither. A benefit-cost ratio that discounts costs and takes benefits at face value will justify almost anything.

Measuring what was actually delivered

EXHIBIT 2

FORMULA	MEANING	DOMAIN
Benefit realised = Actual measure – Baseline measure	Benefits tracking	16.3
Benefits shortfall = Forecast – Realised	Realisation gap	16.3
Benefit dependency: benefit lands only when all enablers deliver	Benefits logic	15.4

Without a pre-delivery baseline, a realisation figure cannot be computed — only asserted. Capture it before delivery starts, not at handover, when the comparison no longer exists. The Institute registers the benefits measure as **EVA(benefit)** deliberately, to keep it clear of **EV** in earned value.

02

Estimating and the network.

Domain 6 — the quantitative flagship. Duration, sequence and float.



PCI AI

| Project Controls
Institute Global, Inc.

Duration under uncertainty

EXHIBIT 3

FORMULA	MEANING	DOMAIN
$tE = (O + 4M + P) \div 6$	PERT expected duration	6.2
$tE = (O + M + P) \div 3$	Triangular expected duration	6.2
$\sigma = (P - O) \div 6$	PERT standard deviation	6.2
$\sigma^2 \text{ path} = \sum \sigma^2 \text{ along the path}$	Path variance	6.4
$\text{Analogous} = \text{Past} \times (\text{this driver} \div \text{past driver})$	Top-down estimating	6.2
$\text{Parametric} = \text{Parameter} \times \text{Rate}$	Rate-based estimating	6.2

Variances add. Standard deviations do not. Sum the σ^2 along the path and take the square root once, at the end. And three points from a single estimator is one number wearing three hats.

The network and its float

EXHIBIT 4

FORMULA	MEANING	DOMAIN
$EF = ES + \text{Duration}$	Forward pass	6.3
$LS = LF - \text{Duration}$	Backward pass	6.3
$TF = LS - ES = LF - EF$	Total float	6.3
$FF = \min(\text{successor } ES) - EF$	Free float	6.3
Critical path: $TF = 0$	The longest path	6.3
$\text{Crash cost slope} = (\text{Crash} - \text{Normal cost}) \div (\text{Normal} - \text{Crash duration})$	Cost per period saved	6.5

Total float from the starts must equal total float from the finishes. If the two forms disagree, one of your passes is wrong. It costs nothing to check and catches a whole class of error — and crashing an activity off the critical path spends money to save nothing.

03

Cost, earned value and risk.

Domains 7 and 8. Where the project stands and where it will end.



PCI AI

| Project Controls
Institute Global, Inc.

Position and forecast

EXHIBIT 5

FORMULA	MEANING	DOMAIN
$CV = EV - AC \cdot SV = EV - PV$	Variances	7.3
$CPI = EV \div AC \cdot SPI = EV \div PV$	Performance indices	7.3
Cost to date = Actuals + Accruals	True cost to date	7.2
$EAC = AC + (BAC - EV)$	The variance was a one-off	7.4
$EAC = BAC \div CPI$	Performance to date persists	7.4
$EAC = AC + (BAC - EV) \div (CPI \times SPI)$	Cost and schedule both persist	7.4
$VAC = BAC - EAC$	Variance at completion	7.4
$TCPI = (BAC - EV) \div (BAC - AC)$	Efficiency required to hit budget	7.4

The third line breaks the first two. If **AC** carries invoiced cost alone, **CPI** overstates performance and **EAC** understates cost — both invisibly. And set **TCPI** beside the **CPI** actually achieved: a **TCPI** far above it asserts a gain nobody has demonstrated.

Schedule performance in time

EXHIBIT 6

FORMULA	MEANING	DOMAIN
ES	Earned schedule — the time at which the EV achieved was planned	6.4
$SV(t) = ES - AT$	Schedule variance in time	6.4
$SPI(t) = ES \div AT$	Time-based schedule efficiency	6.4

SV and SPI are denominated in money. At completion **EV = PV**, so both return to zero and 1.00 however late you finished. The time-based pair does not converge, which makes it the honest late-stage measure.

Risk, priced properly

EXHIBIT 7

FORMULA	MEANING	DOMAIN
EMV = Σ (probability $\times$ impact)	Expected monetary value	8.2
Risk score = Probability rating $\times$ Impact rating	Qualitative sorting only	8.2
Contingency $\approx \Sigma$ EMV, or P80 – P50	Risk-based contingency	8.3
Value of information = EV(with) – EV(without)	Whether to buy analysis	8.4
Residual exposure = Residual probability $\times$ Residual impact	After response	8.3

Ordinal ratings do not multiply. A $3 \times 4 = 12$ heat-map score is a sorting device. It cannot be summed across risks and it cannot size a contingency. For anything touching money, use expected value or simulation.

04

The governance arithmetic.

Original to the PML-AI Body of Knowledge. Governance is the part of delivery everyone manages by instinct — these formulas price it.



PCI AI

| Project Controls
Institute Global, Inc.

What a decision costs you in time

$$E[\text{wait}] = M \div 2 + L$$

Meeting interval **M**, paper lead time **L**. The expected wait for a committee decision — and it **sums across escalation tiers**.

A monthly committee with a two-week paper deadline has an expected wait of four weeks. Escalate through two tiers and it is eight, before anyone has disagreed about anything. Multiply that by the cost of delay and governance stops being free.

The price of elapsed time

Cost of delay = Value forgone per unit of elapsed time

This is the exchange rate between time and money — the price at which governance latency, gate duration and escalation paths are all evaluated.

Without a cost of delay, every governance argument is aesthetic. With one, "add another approval tier" becomes a number, and the conversation changes character entirely.

Is this gate worth holding

Gate net value = $P(\text{defect}) \times \text{build-fix cost} - [\text{review cost} + \text{elapsed} \times \text{cost of delay} + P(\text{defect}) \times (P(\text{detect}) \times \text{design-fix} + P(\text{miss}) \times \text{build-fix})]$

A gate earns its place only when the defects it catches cost more than the gate costs to hold.

A gate that catches nothing is not free — it costs its review effort plus its elapsed time priced at the cost of delay. This is the formula that retires stage gates, and almost no organisation computes it.

Can your governance absorb the demand

Capacity = Meetings per year × Substantive items per meeting
Utilisation = Demand ÷ Capacity

Governance queues behave like any other constrained system. Push utilisation towards 1.0 and the wait time rises non-linearly — which is felt as "the board is a bottleneck" long before anyone measures it.

The architecture choice, priced

$$\text{Mesh interfaces} = n(n - 1) \div 2$$

Layered interfaces = n to an integration layer

Adding one party to a mesh of **n** creates **n** new interfaces, not one.

This is the same arithmetic as communication channels, applied to system and organisational design. It is the case for an integration layer, and it is why adding a party to a late programme so rarely accelerates it.

DOMAIN 4 · THE HUNDRED-PER-CENT RULE

The rule that catches missing scope

$\Sigma \text{ children} - \text{parent} = 0$ at every level of the WBS

A non-zero result is an omission or a duplication. There is no third explanation.

Run it as an arithmetic check, not a review opinion. It is the cheapest scope-assurance test available and it takes one column in a spreadsheet.

Pricing a change honestly

**Assessed total impact = direct +
(schedule weeks × cost of delay) +
rework + interface re-verification +
regression + documentation**

The basis a delegation threshold must read on — not the direct cost alone.

Approve on direct cost and you have set your delegation threshold against a fraction of the real number. Small changes clear the threshold, and the missing terms land later as schedule.

Measuring the drift

Baseline drift = (change count × average direct cost) + (affected count × average weeks × cost of delay)

The cumulative-test threshold is derived from the observed change rate.

Individually approvable, collectively fatal. Every change passed its own test; nobody ran the cumulative one. This formula is the cumulative test.

05

Adaptive delivery and flow.

Domain 13. The same questions, different instruments.



PCI AI

| Project Controls
Institute Global, Inc.

Flow and forecast

EXHIBIT 8

FORMULA	MEANING	DOMAIN
Velocity = Points completed ÷ Sprint	Delivery rate	13.3
Sprints remaining = Points remaining ÷ Average velocity	Forecast to completion	13.3
Capacity = Members × Available days × Focus factor	Sprint capacity	13.3
Cycle time = WIP ÷ Throughput	Little's Law	13.4
Throughput = Items completed ÷ Period	Delivery rate in items	13.4
Flow efficiency = Active time ÷ Elapsed time	Proportion actually worked	13.4
WSJF = Cost of delay ÷ Job size	Prioritisation	13.3

Halve the work in progress and cycle time halves, at the same throughput. That is why limiting WIP speeds delivery without anyone working faster. And **WSJF** requires a real cost of delay — guess it and you have ranked by guess.

Burn-down hides the thing you need

A **burn-down** plots work remaining, so scope added mid-flight looks identical to slow progress.

A **burn-up** plots completed work and total scope as two separate lines.

Scope growth becomes visible as scope growth. For anything reporting to a phase gate, use burn-up. The second line is the entire point.

06

Procurement, quality and portfolio.

Domains 9, 10 and 15.



PCI AI

| Project Controls
Institute Global, Inc.

Where the incentive stops working

EXHIBIT 9

FORMULA	MEANING	DOMAIN
$PTA = [(Ceiling - Target\ price) \div Buyer\ share] + Target\ cost$	Point of total assumption	10.3
$Fee = Target\ fee + Share \times (Target\ cost - Actual\ cost)$	Incentive fee	10.3
$Pain/gain = Share\ ratio \times (Actual - Target)$	Target-cost mechanism	10.3
$LD\ exposure = LD\ rate \times Days\ late$	Liquidated damages	10.4
$Weighted\ score = \Sigma (criterion\ weight \times score)$	Tender evaluation	10.2

Above the PTA the seller absorbs every further pound. Check it in one line: at an actual cost equal to the PTA, the final price must equal the ceiling. If it does not, you have used the seller's share ratio in the denominator instead of the buyer's.

Quality, measured

EXHIBIT 10

FORMULA	MEANING	DOMAIN
Cost of quality = Prevention + Appraisal + Internal + External failure	Total quality cost	9.2
DPMO = (Defects ÷ (Units × Opportunities)) × 1,000,000	Defect rate	9.3
Control limits = Mean ± 3σ	Statistical process control	9.3
Process capability = (USL – LSL) ÷ 6σ	Capability index	9.3
First-pass yield = Passing first time ÷ Started	Right first time	9.3

Control limits describe what the process does. Specification limits describe what the customer requires. A process can be perfectly in control and entirely incapable — and the response to each is different.

Portfolio and capacity

EXHIBIT 11

FORMULA	MEANING	DOMAIN
Portfolio value = Σ NPV of components	Aggregate value	15.2
Scoring model = Σ (weight $\times$ score)	Prioritisation	15.2
Σ resource demand $\leq$ Available capacity	Feasibility constraint	15.3
Resource utilisation = $\frac{\text{Productive}}{\text{Available hours}}$	Utilisation	7.5

Do not plan to 100% utilisation. It leaves nothing to absorb variability, and queues grow non-linearly as utilisation approaches full — the same effect as committee capacity in section 04, in a different queue.

MOST MISAPPLIED · 1-5

The ten that cost the most

1. Risk scores multiplied and then summed
2. **SV** read as a schedule measure late in a project
3. **PTA** computed with the seller's share ratio
4. Velocity taken as best-ever rather than a rolling average
5. Burn-down used for gate reporting

MOST MISAPPLIED · 6-10

And the other five

1. Standard deviations summed along a path
2. Crashing an activity that is not on the critical path
3. Control limits confused with specification limits
4. Change approved on direct cost alone, not assessed total impact
5. Benefits claimed with no pre-delivery baseline

Nine of these produce a report that passes review. The tenth produces a benefits case that cannot be checked at all.



PCI AI

| Project Controls
Institute Global, Inc.

The Body of Knowledge is open.

Every formula here is developed to full depth in the PML-AI Body of Knowledge — sixteen domains with worked examples and calculation exercises, at the domain cited on each slide. The governance arithmetic in section 04 is original to it.

Published openly. **No registration, no email gate.**

projectcontrolsinstitute.org